

Grade 11 – Term 1

C2 Practice Activity 2 with Solutions

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Evaluated criterion

- C2: Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.

1 Introduction

Problems:

- 1.1. Uniform distribution
 - 1.2. Linear distribution
 - 1.3. Triangular distribution
 - 1.4. Piecewise uniform distribution
 - 1.5. Piecewise linear distribution
-

For every problem, use the same seven-step routine: (1) identify the **support**; (2) name the density's shape; (3) use total area = 1 to find any unknown height or parameter; (4) mark the requested interval; (5) split its region into familiar figures; (6) calculate and add or subtract their areas; and (7) state the probability in context. A density is a height, so it may exceed 1; it is the *area* that must total 1. Endpoints do not affect a continuous probability, for example $P(a < X < b) = P(a \leq X \leq b)$.

1.1 Uniform distribution

A uniform density has one constant height on support $[L, U]$:

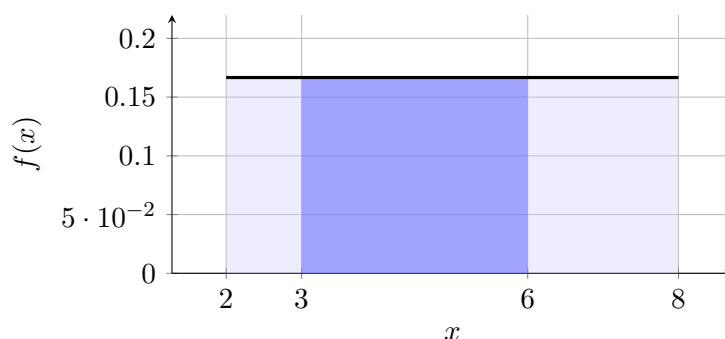
$$f_X(x) = \begin{cases} h, & L \leq x \leq U, \\ 0, & \text{otherwise.} \end{cases} \quad (U - L)h = 1, \quad h = \frac{1}{U - L}.$$

Thus every probability is a rectangle: $P(a \leq X \leq b) = (b - a)h$ for $L \leq a < b \leq U$.

Worked example. A delivery time T is equally plausible from 2 to 8 minutes. The support is $[2, 8]$ and the whole rectangle has width 6 and height h , so $6h = 1$ and $h = 1/6$. For $3 \leq T \leq 6$, the relevant rectangle has width $6 - 3 = 3$ and height $1/6$:

$$P(3 \leq T \leq 6) = A_{\text{rectangle}} = 3 \left(\frac{1}{6} \right) = \frac{1}{2}.$$

There is a 50% chance that delivery takes from 3 to 6 minutes.



1.2 Linear distribution

A single linear density has $f_X(x) = mx + d$ throughout one support $[L, U]$ (and 0 outside), with nonnegative endpoint heights $h_L = f(L)$ and $h_U = f(U)$. Its whole area is the trapezoid

$$1 = \frac{1}{2}(h_L + h_U)(U - L).$$

An interval $[a, b]$ is also a trapezoid of width $b - a$ and parallel heights $f(a), f(b)$. A triangle is only the special case in which one endpoint height is zero.

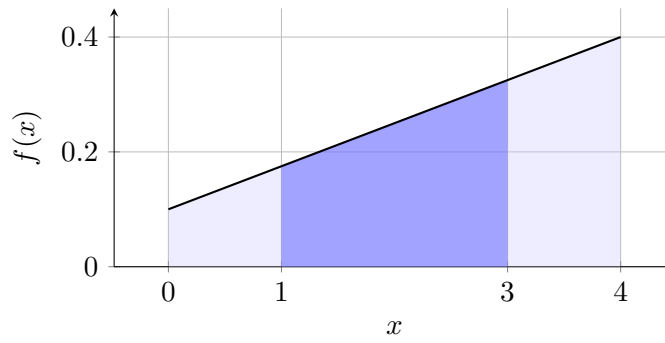
Worked example. The density of a shop's daily price adjustment X (in dollars) is $f(x) = kx + 0.1$ on $[0, 4]$. The support width is 4; its endpoint heights are 0.1 and $4k + 0.1$. Normalisation gives

$$1 = \frac{1}{2}[0.1 + (4k + 0.1)]4 = 8k + 0.4, \quad k = 0.075.$$

For $1 \leq X \leq 3$, $f(1) = 0.175$ and $f(3) = 0.325$. The required figure is a trapezoid of width 2 with those parallel heights:

$$P(1 \leq X \leq 3) = \frac{1}{2}(0.175 + 0.325)(2) = 0.5.$$

Both support endpoints have positive density, so this is genuinely linear, not triangular.



1.3 Triangular distribution

A triangular density has support $[L, U]$, height 0 at both ends, and mode M ($L < M < U$) with height H . It rises linearly on $[L, M]$ and falls linearly on $[M, U]$. Since the whole figure has base $U - L$,

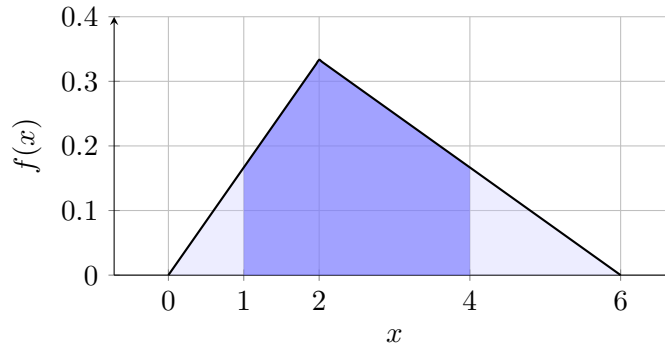
$$1 = \frac{1}{2}(U - L)H, \quad H = \frac{2}{U - L}.$$

Heights inside either sloping side are found by proportional similar triangles.

Worked example. A project's cost overrun C (thousand dollars) has minimum 0, most likely value 2, and maximum 6. Its support is $[0, 6]$, mode is 2, and $\frac{1}{2}(6)H = 1$, so $H = 1/3$. To find $P(1 \leq C \leq 4)$, subtract two tail triangles. At $C = 1$ the height is $1/6$; at $C = 4$ it is also $1/6$. The left tail has base 1, height $1/6$; the right tail has base 2, height $1/6$:

$$P(1 \leq C \leq 4) = 1 - \frac{1}{2}(1)\left(\frac{1}{6}\right) - \frac{1}{2}(2)\left(\frac{1}{6}\right) = 1 - \frac{1}{12} - \frac{1}{6} = \frac{3}{4}.$$

The region crosses the mode, but subtraction avoids splitting it into two trapezoids.



1.4 Piecewise uniform distribution

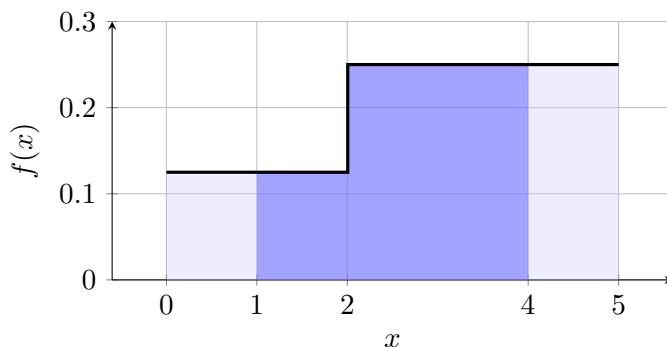
Here the density is constant *within* each interval but may jump at a breakpoint. Normalisation is the sum of rectangles: $1 = \sum(\text{segment width})(\text{segment height})$.

Worked example. A warehouse waiting time W has

$$f_W(w) = \begin{cases} k, & 0 \leq w < 2, \\ 2k, & 2 \leq w \leq 5, \\ 0, & \text{otherwise.} \end{cases}$$

The support is $[0, 5]$. A width-2, height- k rectangle plus a width-3, height- $2k$ rectangle must have area 1: $2k + 6k = 1$, so $k = 1/8$. The event $1 \leq W \leq 4$ spans the jump and uses two rectangles:

$$P(1 \leq W \leq 4) = \underbrace{(2-1)(1/8)}_{A_1} + \underbrace{(4-2)(1/4)}_{A_2} = \frac{1}{8} + \frac{1}{2} = \frac{5}{8}.$$



1.5 Piecewise linear distribution

A piecewise-linear density joins straight segments at stated breakpoints. Read the height at each endpoint and breakpoint, then treat each piece as a triangle, rectangle, or trapezoid. Continuity means adjacent pieces meet at the same height.

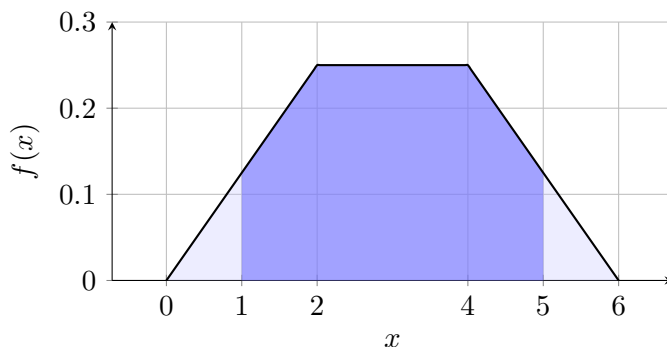
Worked example. Demand D has support $[0, 6]$. Its density joins $(0, 0)$, $(2, H)$, $(4, H)$, and $(6, 0)$. The whole shape is a left triangle (base 2), rectangle (width 2), and right triangle (base 2):

$$1 = \frac{1}{2}(2)H + (2)H + \frac{1}{2}(2)H = 4H, \quad H = \frac{1}{4}.$$

For $1 \leq D \leq 5$, subtract the two congruent tail triangles. By linearity, the heights at 1 and 5 are $H/2 = 1/8$:

$$P(1 \leq D \leq 5) = 1 - \frac{1}{2}(1)(\frac{1}{8}) - \frac{1}{2}(1)(\frac{1}{8}) = \frac{7}{8}.$$

This reusable approach identifies all three geometric pieces and uses subtraction from total area 1.



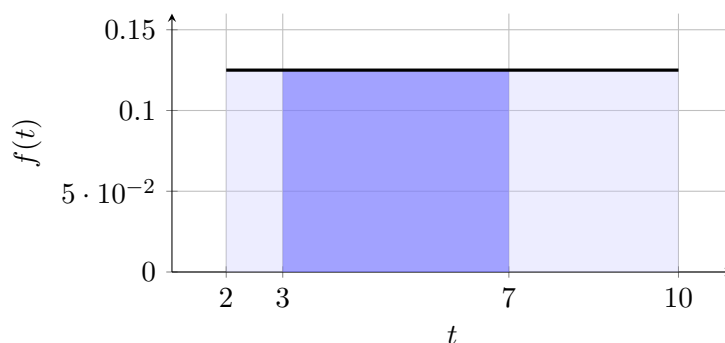
2 Uniform Distribution

Problems:

2.1. Microcredit disbursement window

2.1 Microcredit disbursement window

A Bogotá fintech releases approved working-capital loans to neighbourhood businesses after completing fraud and bank-account checks. Let T be the number of hours from approval to disbursement. For applications that require no manual review, operational data support a uniform model from 2 to 10 hours. A bakery owner needs the money within seven hours to pay a flour supplier before dispatch closes; the fintech also monitors a four-hour service band when planning its automated processing capacity.



Questions/Tasks.

C2.1 Identify the model, random variable, parameters, and support; then write its pdf.

C2.2 Calculate $P(3 \leq T \leq 7)$ from the relevant rectangular area.

C2.3 Use a complement to calculate $P(T > 7)$.

C2.4 Interpret both probabilities for the fintech and the bakery owner.

C2.1 - Identify the model, random variable, parameters, and support; then write its pdf.

The random variable T measures disbursement time in hours. The model is uniform with lower parameter $L = 2$, upper parameter $U = 10$, and support $[2, 10]$. If its constant height is h , the entire pdf is a rectangle of width $10 - 2 = 8$. Since total probability is 1,

$$1 = A_{\text{rectangle}} = (10 - 2)h = 8h \implies h = \frac{1}{8}.$$

Therefore

$$f_T(t) = \begin{cases} \frac{1}{8}, & 2 \leq t \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

The equation and graph agree: the pdf is constant exactly on the stated operational window, and its full rectangular area is $8(1/8) = 1$.

C2.2 - Calculate $P(3 \leq T \leq 7)$ from the relevant rectangular area.

The shaded service band has width $7 - 3 = 4$ and height $1/8$. Thus

$$P(3 \leq T \leq 7) = A_{\text{rectangle}} = (7 - 3) \left(\frac{1}{8} \right) = \frac{4}{8} = \frac{1}{2}.$$

Hence half of comparable automatic disbursements take between three and seven hours.

C2.3 - Use a complement to calculate $P(T > 7)$.

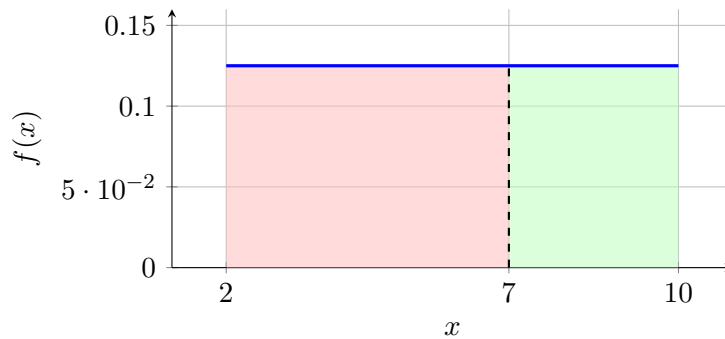
The event $T \leq 7$ is the rectangle from 2 to 7:

$$P(T \leq 7) = (7 - 2) \left(\frac{1}{8} \right) = \frac{5}{8}.$$

Because an exact time has probability zero in a continuous model,

$$P(T > 7) = 1 - P(T \leq 7) = 1 - \frac{5}{8} = \frac{3}{8}.$$

This is the same as the right-hand rectangle $(10 - 7)(1/8) = 3/8$, which checks the complement calculation.



C2.4 - Interpret both probabilities for the fintech and the bakery owner.

The value $P(T > 7) = 3/8 = 37.5\%$ means the bakery owner faces a substantial risk of missing the supplier's dispatch if relying on this loan alone. Meanwhile, $P(3 \leq T \leq 7) = 1/2$ tells the fintech that half of these cases occupy its central four-hour processing band. These figures can support a warning to the borrower and the allocation of extra automated-processing capacity; they do not guarantee the outcome of one application.



3 Linear Distribution

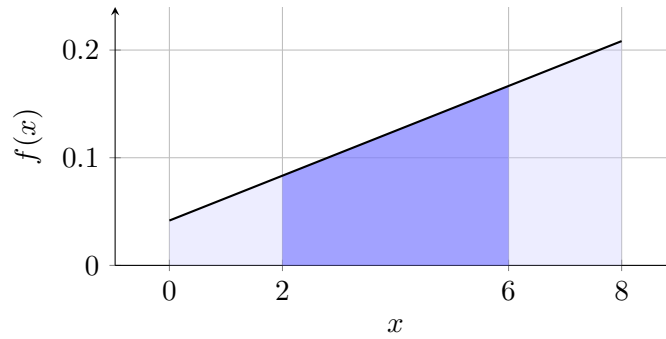
Problems:

3.1. Coffee auction premium

3.1 Coffee auction premium

A Bogotá specialty-coffee roaster buys small lots from Colombian producers. On an auction morning, let X be the premium above the reference price, measured in thousands of pesos per kilogram. The premium lies between 0 and 8,

and buyer competition makes larger premiums progressively more plausible. The purchasing manager proposes the single linear density $f_X(x) = k(x + 2)$ on that interval. Premiums above 6 thousand pesos threaten the margin on existing café contracts, while the 2–6 band can be absorbed without changing menu prices.



Questions/Tasks.

C2.1 Identify the model and support, determine k geometrically, and write the pdf.

C2.2 Calculate $P(2 \leq X \leq 6)$ as a trapezoid.

C2.3 Calculate $P(X > 6)$ and verify it using a complement.

C2.4 Interpret the probabilities for the purchasing decision.

C2.1 - Identify the model and support, determine k geometrically, and write the pdf.

This is a linear distribution, not a triangular distribution: its support is $[0, 8]$ and both endpoint heights are positive,

$$f_X(0) = 2k, \quad f_X(8) = 10k.$$

The full region is a trapezoid of width 8. Normalisation therefore gives

$$1 = \frac{1}{2}(8)(2k + 10k) = 48k \implies k = \frac{1}{48}.$$

Consequently,

$$f_X(x) = \begin{cases} \frac{x+2}{48}, & 0 \leq x \leq 8, \\ 0, & \text{otherwise.} \end{cases}$$

The increasing line represents the auction assumption that a higher premium has a greater density, while the total trapezoidal area confirms a valid pdf.

C2.2 - Calculate $P(2 \leq X \leq 6)$ as a trapezoid.

Read the two heights from the equation:

$$f_X(2) = \frac{4}{48} = \frac{1}{12}, \quad f_X(6) = \frac{8}{48} = \frac{1}{6}.$$

The shaded region is a width-4 trapezoid, not a rectangle, because its height changes across the interval. Hence

$$\begin{aligned} P(2 \leq X \leq 6) &= \frac{1}{2}(6-2)(f_X(2) + f_X(6)) \\ &= \frac{1}{2}(4)\left(\frac{1}{12} + \frac{1}{6}\right) = 2\left(\frac{3}{12}\right) = \frac{1}{2}. \end{aligned}$$

Thus the premium falls in the absorbable band in 50% of auctions described by this model.

C2.3 - Calculate $P(X > 6)$ and verify it using a complement.

Directly, the region from 6 to 8 is a trapezoid with heights $f_X(6) = 1/6$ and $f_X(8) = 10/48 = 5/24$:

$$P(X > 6) = \frac{1}{2}(8 - 6) \left(\frac{1}{6} + \frac{5}{24} \right) = \frac{4}{24} + \frac{5}{24} = \frac{3}{8}.$$

For a check, the region from 0 to 6 has heights $1/24$ and $1/6$:

$$P(X \leq 6) = \frac{1}{2}(6) \left(\frac{1}{24} + \frac{1}{6} \right) = \frac{5}{8}, \quad 1 - \frac{5}{8} = \frac{3}{8}.$$

Both geometric routes agree.

C2.4 - Interpret the probabilities for the purchasing decision.

Although half of auctions land in the manageable 2–6 band, the 37.5% chance of a premium above 6 is the more important margin-risk measure. The manager could cap bids, renegotiate café prices, or reserve a smaller lot; the probability quantifies the frequency of the risky region, not the size of a loss.



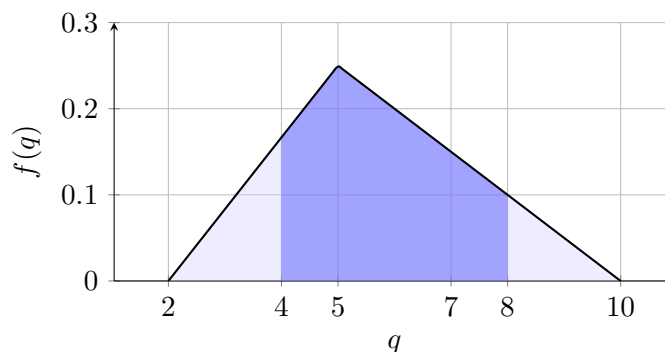
4 Triangular Distribution

Problems:

4.1. Last-mile order volume

4.1 Last-mile order volume

A Bogotá last-mile cooperative must contract motorcycles before a national online-sales event. Let Q be next-day order volume in thousands of orders. Planners set an optimistic minimum of 2, a most likely value of 5, and a capacity-limited maximum of 10, so they use a triangular model. The regular fleet handles 4–8 thousand orders; demand above 7 thousand triggers an expensive same-day contract with external couriers.



Questions/Tasks.

C2.1 Interpret the parameters, find the peak height, and write the piecewise pdf.

C2.2 Find $P(4 \leq Q \leq 8)$ by subtracting two tail triangles.

C2.3 Find $P(Q > 7)$ from a geometric area.

C2.4 Interpret both probabilities for fleet contracting.

C2.1 - Interpret the parameters, find the peak height, and write the piecewise pdf.

The support endpoints $L = 2$ and $U = 10$ are the modelled minimum and maximum; the mode $M = 5$ is the most likely order volume. If the peak is H , the whole triangle has base $10 - 2 = 8$, so

$$1 = \frac{1}{2}(8)H \implies H = \frac{1}{4}.$$

The rising segment has slope $(1/4)/(5 - 2) = 1/12$, and the falling segment has slope $-(1/4)/(10 - 5) = -1/20$. Therefore

$$f_Q(q) = \begin{cases} \frac{q-2}{12}, & 2 \leq q \leq 5, \\ \frac{10-q}{20}, & 5 < q \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

Both formulas give $f_Q(5) = 1/4$, so the equation connects continuously at the mode and matches the displayed triangle.

C2.2 - Find $P(4 \leq Q \leq 8)$ by subtracting two tail triangles.

The requested interval crosses the mode. Rather than add two trapezoids, subtract the two simpler tails. From the pdf,

$$f_Q(4) = \frac{4-2}{12} = \frac{1}{6}, \quad f_Q(8) = \frac{10-8}{20} = \frac{1}{10}.$$

The left tail from 2 to 4 and right tail from 8 to 10 are triangles:

$$A_L = \frac{1}{2}(4-2)\frac{1}{6} = \frac{1}{6}, \quad A_R = \frac{1}{2}(10-8)\frac{1}{10} = \frac{1}{10}.$$

Thus

$$P(4 \leq Q \leq 8) = 1 - A_L - A_R = 1 - \frac{1}{6} - \frac{1}{10} = \frac{11}{15}.$$

C2.3 - Find $P(Q > 7)$ from a geometric area.

The point 7 lies on the descending side, where

$$f_Q(7) = \frac{10-7}{20} = \frac{3}{20}.$$

The complete region to its right is a triangle of base $10 - 7 = 3$ and height $3/20$. Therefore

$$P(Q > 7) = \frac{1}{2}(3)\left(\frac{3}{20}\right) = \frac{9}{40} = 22.5\%.$$

This tail calculation uses the graph directly and needs no integration.

C2.4 - Interpret both probabilities for fleet contracting.

The regular fleet range covers order volume with probability $11/15 \approx 73.3\%$. However, external couriers are needed with probability $9/40 = 22.5\%$. The cooperative can use the first figure to evaluate its base capacity and the second to negotiate contingent motorcycle availability; the remaining probability is not automatically a failure because volumes below 4 require fewer riders.



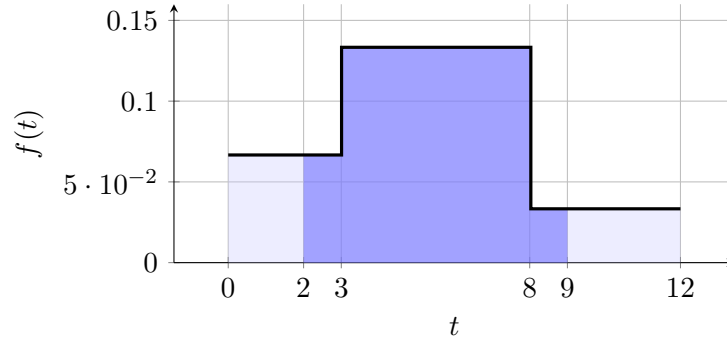
5 Piecewise Uniform Distribution

Problems:

5.1. Merchant reconciliation time

5.1 Merchant reconciliation time

A Colombian payments processor reconciles QR-code transactions from Bogotá merchants before releasing daily balances. Let T be reconciliation time in minutes. Monitoring suggests three operational regimes: density k from 0 to 3 minutes, $2k$ from 3 to 8 minutes, and $k/2$ from 8 to 12 minutes. The processor promises that most balances will clear from minute 2 through minute 9; times above 8 minutes require a slower exception queue and increase support contacts.



Questions/Tasks.

- C2.1** Identify the support and regimes, determine k , and write the pdf.
- C2.2** Find $P(2 \leq T \leq 9)$ by combining three rectangles.
- C2.3** Find $P(T > 8)$ and check it with a complement.
- C2.4** Interpret the results for the service promise and exception queue.

C2.1 - Identify the support and regimes, determine k , and write the pdf.

The support is $[0, 12]$. The model is piecewise uniform because the pdf is constant within each regime but jumps at 3 and 8. The whole graph is the sum of three rectangles:

$$1 = \underbrace{3k}_{0-3} + \underbrace{(8-3)(2k)}_{3-8} + \underbrace{(12-8)\frac{k}{2}}_{8-12} = 3k + 10k + 2k = 15k.$$

Hence $k = 1/15$, and

$$f_T(t) = \begin{cases} \frac{1}{15}, & 0 \leq t < 3, \\ \frac{2}{15}, & 3 \leq t < 8, \\ \frac{1}{30}, & 8 \leq t \leq 12, \\ 0, & \text{otherwise.} \end{cases}$$

The three rectangular areas sum to 1, so this nonnegative piecewise function is a valid pdf.

C2.2 - Find $P(2 \leq T \leq 9)$ by combining three rectangles.

The interval crosses both breakpoints. Split it into $[2, 3)$, $[3, 8)$, and $[8, 9]$ and use the height of each regime:

$$\begin{aligned} P(2 \leq T \leq 9) &= \underbrace{(3-2)\frac{1}{15}}_{A_1} + \underbrace{(8-3)\frac{2}{15}}_{A_2} + \underbrace{(9-8)\frac{1}{30}}_{A_3} \\ &= \frac{1}{15} + \frac{10}{15} + \frac{1}{30} = \frac{2+20+1}{30} = \frac{23}{30}. \end{aligned}$$

Thus about 76.7% of reconciliations fall inside the stated service window.

C2.3 - Find $P(T > 8)$ and check it with a complement.

Directly, the exception-queue region is a width-4 rectangle of height $1/30$:

$$P(T > 8) = (12 - 8)\frac{1}{30} = \frac{2}{15}.$$

For the complement, the first two full rectangles have area

$$P(T \leq 8) = 3 \left(\frac{1}{15} \right) + 5 \left(\frac{2}{15} \right) = \frac{1}{5} + \frac{2}{3} = \frac{13}{15},$$

so $1 - 13/15 = 2/15$, confirming the direct calculation. Whether the endpoint 8 is included is immaterial for this continuous variable.

C2.4 - Interpret the results for the service promise and exception queue.

The proposed 2–9 minute promise contains about 76.7% of cases, so it would be misleading if presented as a near-certain outcome. The exception queue receives $2/15 \approx 13.3\%$ of reconciliations. These probabilities help the processor set an honest merchant message and staff the slower queue, while the model’s regime jumps reflect process categories rather than sudden probability at an exact time.



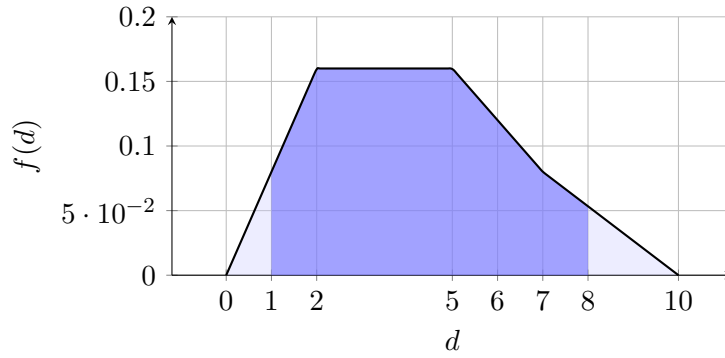
6 Piecewise Linear Distribution

Problems:

6.1. Weekly payout liquidity

6.1 Weekly payout liquidity

A Bogotá marketplace must hold cash for Friday seller payouts, refunds, and chargebacks. Let D be required liquidity in billions of pesos. Its finance team models a continuous piecewise-linear pdf joining $(0, 0)$, $(2, H)$, $(5, H)$, $(7, H/2)$, and $(10, 0)$. The shape represents a ramp-up to common payout requirements, a plateau, and then progressively less likely high-liquidity weeks. The treasury plan covers 1 through 8 billion pesos; requirements above 6 billion trigger a more expensive backup credit line.



Questions/Tasks.

C2.1 Determine H from the component areas and write the piecewise pdf.

C2.2 Find $P(1 \leq D \leq 8)$ by subtracting the two tail triangles.

C2.3 Find $P(D > 6)$ by combining regions on both sides of a breakpoint.

C2.4 Interpret the probabilities for treasury planning.

C2.1 - Determine H from the component areas and write the piecewise pdf.

The support $[0, 10]$ contains four familiar regions: a base-2 triangle, a width-3 rectangle, a width-2 trapezoid with heights H and $H/2$, and a base-3 triangle of height $H/2$. Total area 1 gives

$$\begin{aligned} 1 &= \frac{1}{2}(2)H + 3H + \frac{1}{2}(2) \left(H + \frac{H}{2} \right) + \frac{1}{2}(3) \frac{H}{2} \\ &= H + 3H + \frac{3}{2}H + \frac{3}{4}H = \frac{25}{4}H, \end{aligned}$$

so $H = 4/25$. Equations of the joined segments then give

$$f_D(d) = \begin{cases} \frac{2d}{25}, & 0 \leq d \leq 2, \\ \frac{4}{25}, & 2 < d \leq 5, \\ \frac{9-d}{25}, & 5 < d \leq 7, \\ \frac{2(10-d)}{75}, & 7 < d \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

At $d = 2, 5, 7$ the adjacent formulas agree, so the equation is continuous and matches the graph.

C2.2 - Find $P(1 \leq D \leq 8)$ by subtracting the two tail triangles.

At $d = 1$, $f_D(1) = 2/25$. The left tail is a triangle with base 1:

$$A_L = \frac{1}{2}(1) \frac{2}{25} = \frac{1}{25}.$$

On the last segment, $f_D(8) = 2(10-8)/75 = 4/75$. The right tail from 8 to 10 is another triangle:

$$A_R = \frac{1}{2}(10-8) \frac{4}{75} = \frac{4}{75}.$$

Therefore

$$P(1 \leq D \leq 8) = 1 - A_L - A_R = 1 - \frac{1}{25} - \frac{4}{75} = \frac{68}{75} \approx 90.7\%.$$

The complement method connects the two tail triangles in the graph to the central treasury range without adding four internal pieces.

C2.3 - Find $P(D > 6)$ by combining regions on both sides of a breakpoint.

The event begins inside the segment from 5 to 7 and continues through the segment from 7 to 10. First find the height at 6:

$$f_D(6) = \frac{9-6}{25} = \frac{3}{25}, \quad f_D(7) = \frac{2}{25}.$$

From 6 to 7 the area is a trapezoid; from 7 to 10 it is a triangle:

$$\begin{aligned} P(D > 6) &= \underbrace{\frac{1}{2}(7-6) \left(\frac{3}{25} + \frac{2}{25} \right)}_{A_1} + \underbrace{\frac{1}{2}(10-7) \frac{2}{25}}_{A_2} \\ &= \frac{1}{10} + \frac{3}{25} = \frac{5+6}{50} = \frac{11}{50} = 22\%. \end{aligned}$$

This calculation must include both regions; stopping at the breakpoint 7 would omit high-liquidity outcomes.

C2.4 - Interpret the probabilities for treasury planning.

The standard 1–8 billion peso plan covers approximately 90.7% of weeks. Nevertheless, the backup credit line is triggered in 22% of weeks because its threshold begins at 6, inside that standard range. The two events overlap, so their probabilities should not be added. Treasury can use the 22% figure to price standby funding and the 90.7% figure to evaluate the overall coverage of its cash plan.

